Exploratory Data Analysis

Course: Computational Social Science, Homework #2
Team: FightClub
Research topic: Brainrot analysis
Lecturer: Andrew Kurochkin
Date: October 2026
Place: National University of Kyiv-Mohyla Academy, Faculty of Informatics

Time spent

Name	Time
Danylo	15h 5m
Ivan	12h
Daryna	7h
Total	34h 5m

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0. Overview

Since our goal is many-sided and has certain specific characteristics, we will follow our own flow with exploration and analysis of data. We already have hypotheses to test and we will try to find approaches in order to prove those hypothese right or wrong. We will not follow guidelines or try to check any boxes, instead we will follow our curiosity and play until we have the answer.

However, there is one guideline: before freestyling with analysis - explore the data and brainstorm what visualizations should be made.

Since we have 4 different hypotheses to test, the research is structured into 4 independent parts, each of them testing one hypothesis and doing separate data overview. Each hypothesis is tested with separate datasets and those datasets are reviewed independently in each of 4 parts. After that some of the results and findings are merged.

0.1 Hypotheses

Our topic is analysis of impact of doomscrolling and brainrot on people, discourse, their wellbeing and cognitive abilities over time. We study the effects of doomscrolling with the hypotheses below.

- H1: Discourse is getting shorter and simpler
  - Lexical & NLP approach
  - Temporal/Behavioral aspect
- H2: Short-form platforms accelerated the degradation of discourse
- H3: Wellbeing is getting worse among doomscrollers
- H4: Crises intensify doomscrolling and negativity

0.2 Datasets

Each hypothesis is proven by using different datasets.

H1: Discourse is getting shorter and simpler

Comments and submissions from reddit, collected from 01.2012 to 09.2026, divided into different categories.

1. Short-form: r/memes, r/teenagers
2. Long-form: r/books, r/explainlikeimfive
3. Baseline: r/todayilearned
4. Qualitative: r/nosurf

H2: Short-form platforms accelerated the degradation of discourse

1. Reels/Shorts Consumption vs Attention Span

H3: Wellbeing is getting worse among doomscrollers

1. Social Media Addiction dataset
2. The Dark At The End Of The Tunnel: Doomscrolling On Social Media Newsfeeds
3. Sleep & Doomscrolling Habits Dataset
4. Social Media and Mental Health
5. Student Social Media & Mental Health

H4: Crises intensify doomscrolling and negativity

0.3 Additional approaches

- Questionnaire - work in progress
- YouTube comments - work in progress

1 H1: Discourse is getting shorter and simpler

In this section we test this hypothesis with data. We'll be analyzing the discourse of a dataset of 9.2 million comments and 1.8 million submissions that were posted on reddit between 2012 and 2026.

Before getting into data and into analysis, in order to pick the correct analysis path, let's take a bird's-eye view of our data and on our goal. We want to verify that over the 14 years discourse has been getting shorter and simpler. In order to check that, we have two approaches:

1. **Lexical & NLP approach**
2. **Temporal/Behavioral aspect**

In **Lexical & NLP approach** we analyse sentences and check the following metrics:

1. Vocabulary is shrinking
2. Sentences are getting shorter
    1. Words per comment
    2. Words per sentence
3. Lempel-Ziv algorithm
    1. Concatenation of comments into fixed-size chunks per subreddit per month

In **Temporal/Behavioral aspect** we analyze conversantions and look into:

1. Thread depth
    1. Thread size
    2. Share or replies vs top-level comments
    3. True depth

Ideas for analysis:

1. Readability (Flesch-Kincaid)
2. Avg word length

1.1. Overview

Our dataset consists of 9,237,205 comments and 1,870,257 posts, collected from 01.2012 to 09.2026, from these subreddits:

1. Short-form: r/memes, r/teenagers
2. Long-form: r/books, r/explainlikeimfive
3. Baseline: r/todayilearned
4. Qualitative: r/nosurf

Dataset was downloaded from Arctic Shift API. We didn't download full dataset of comments and posts because of the size and downloading time. Instead, we sampled 10,000 comments per subreddit per month and 2,000 posts per subreddit per month. Samples are downloaded from random time slots. Each month is cut into equal time slots, each slot holding roughly 1,000 comments or 200 posts. For our subreddits total amount of comments is 346.4 million and total amount of posts is 27.3 million. Our sample covers about 2.7% of all comments and 6.9% of all posts.

type	records	subreddits	first_record	last_record	months	files	size_MB
comments	9237205	6	2012-01-01	2026-09-30	177	1062	897.5
posts	1870257	6	2012-01-01	2026-09-30	177	1062	225.4

	comments					\
subreddit	sampled	total	first_month	last_month	coverage_%	
books	1876751	13137631	2012-01	2026-09	14.3	
explainlikeimfive	1852536	21058471	2012-01	2026-09	8.8	
memes	1334537	108528872	2012-01	2026-09	1.2	
nosurf	478540	478745	2012-01	2026-09	100.0	
teenagers	1843775	133272831	2012-01	2026-09	1.4	
todayilearned	1851066	69926671	2012-01	2026-09	2.6	

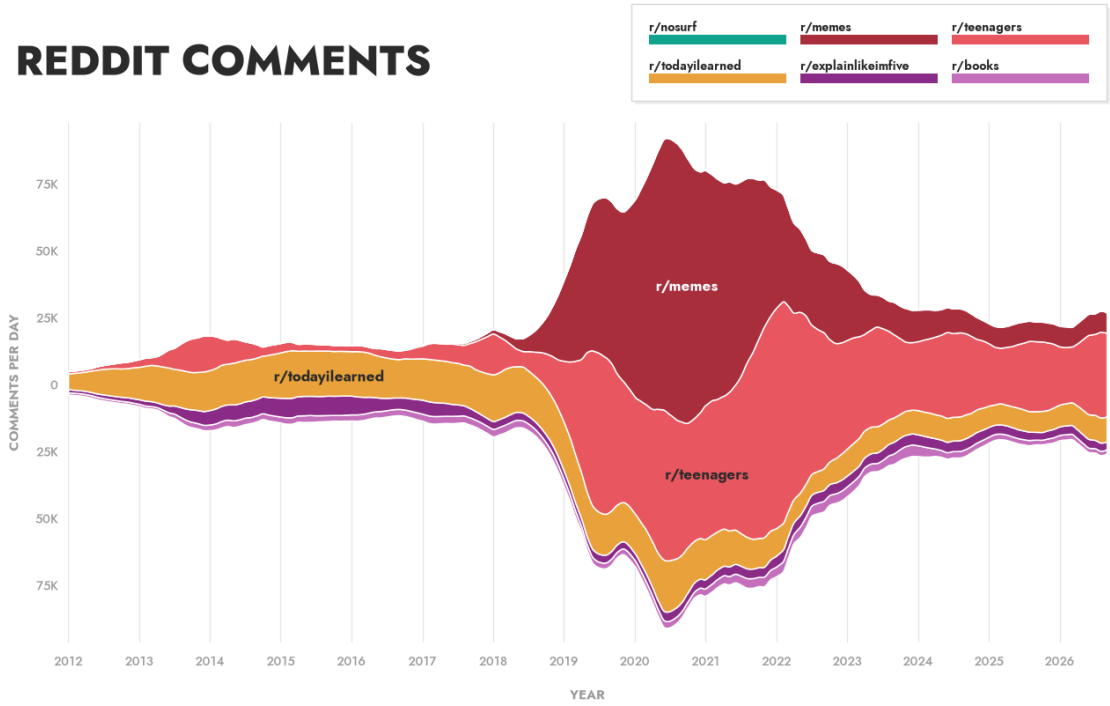
		posts				\
subreddit	complete_months	sampled	total	first_month	last_month	
books	0	368255.0	890216	2012-01	2026-09	
explainlikeimfive	0	369085.0	2000253	2012-01	2026-09	
memes	62	343402.0	11858295	2012-01	2026-09	
nosurf	177	56057.0	56082	2012-01	2026-09	
teenagers	0	363975.0	10305407	2012-01	2026-09	
todayilearned	0	369483.0	2162300	2012-01	2026-09	

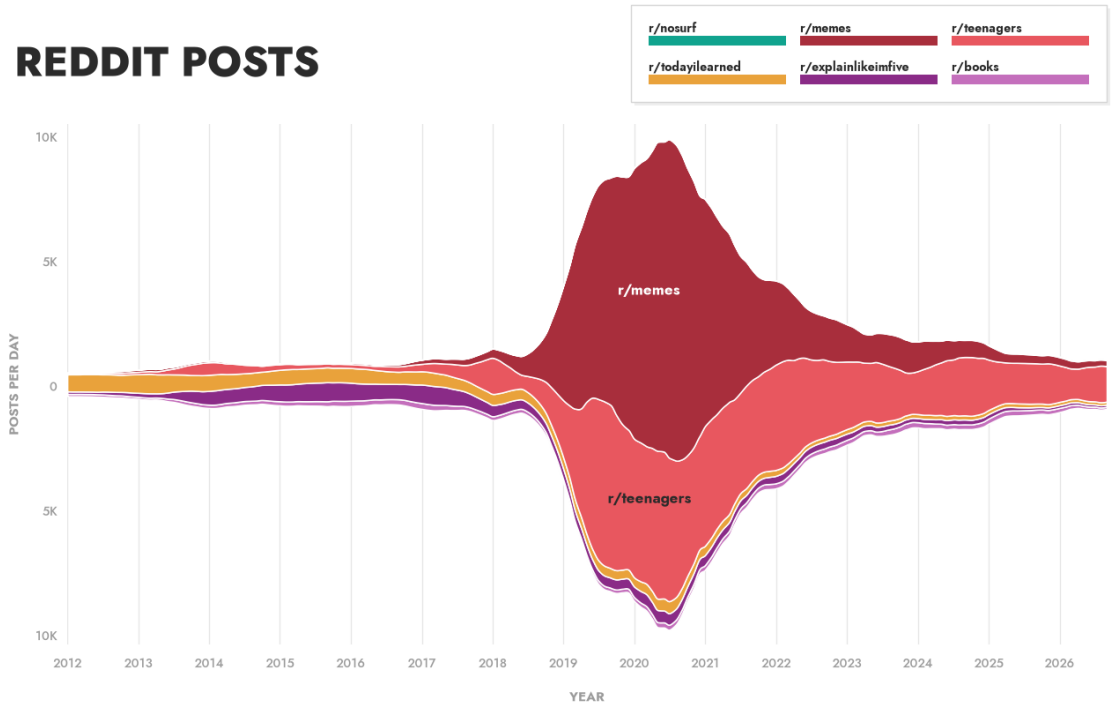
	coverage_%	complete_months

subreddit		
books	41.4	0
explainlikeimfive	18.5	0
memes	2.9	38
nosurf	100.0	177
teenagers	3.5	8
todayilearned	17.1	0

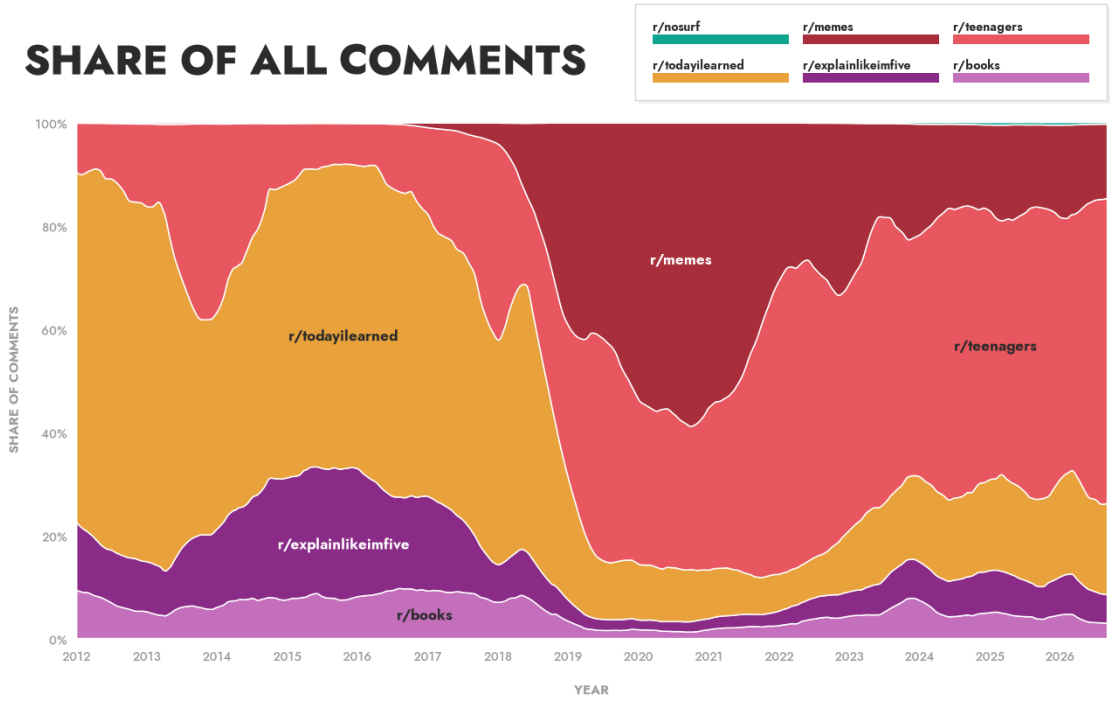
1.2 Activity and distribution

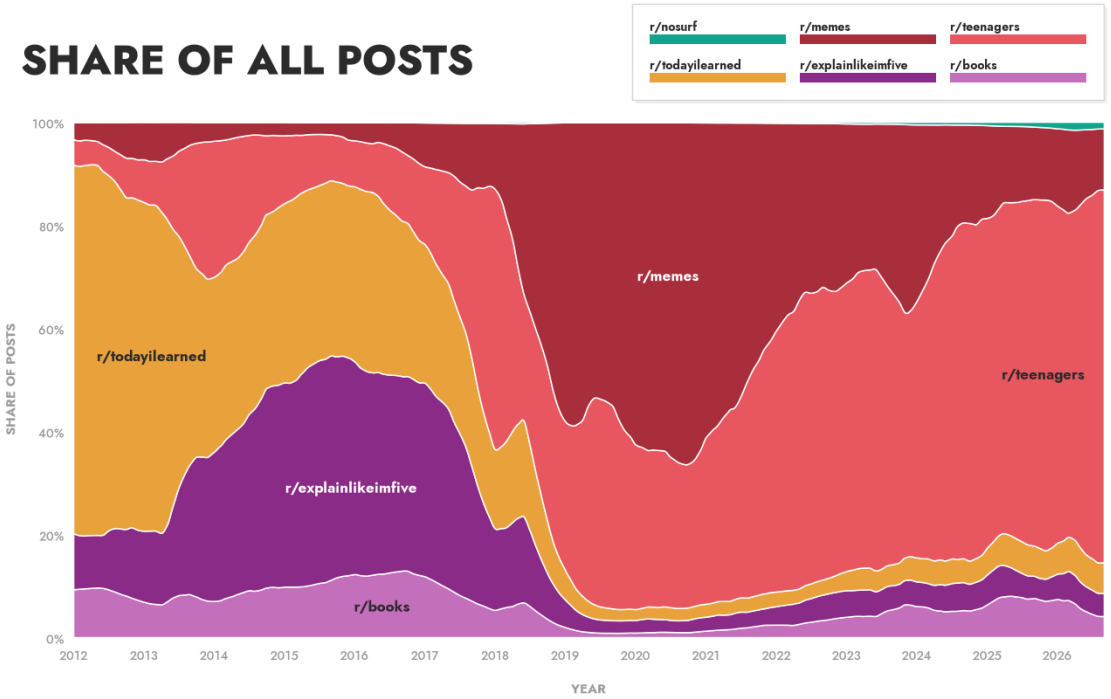
Before analysing the datasets, let's explore the activity over the last 14 years. First 2 plots represent the streamgraphs of comments and posts per day for all subreddit in absolute values. In August 2018 TikTok global, in March 2020 COVID-19 took place and TikTok went viral during the same period of time, as can seen on the streamgraph.



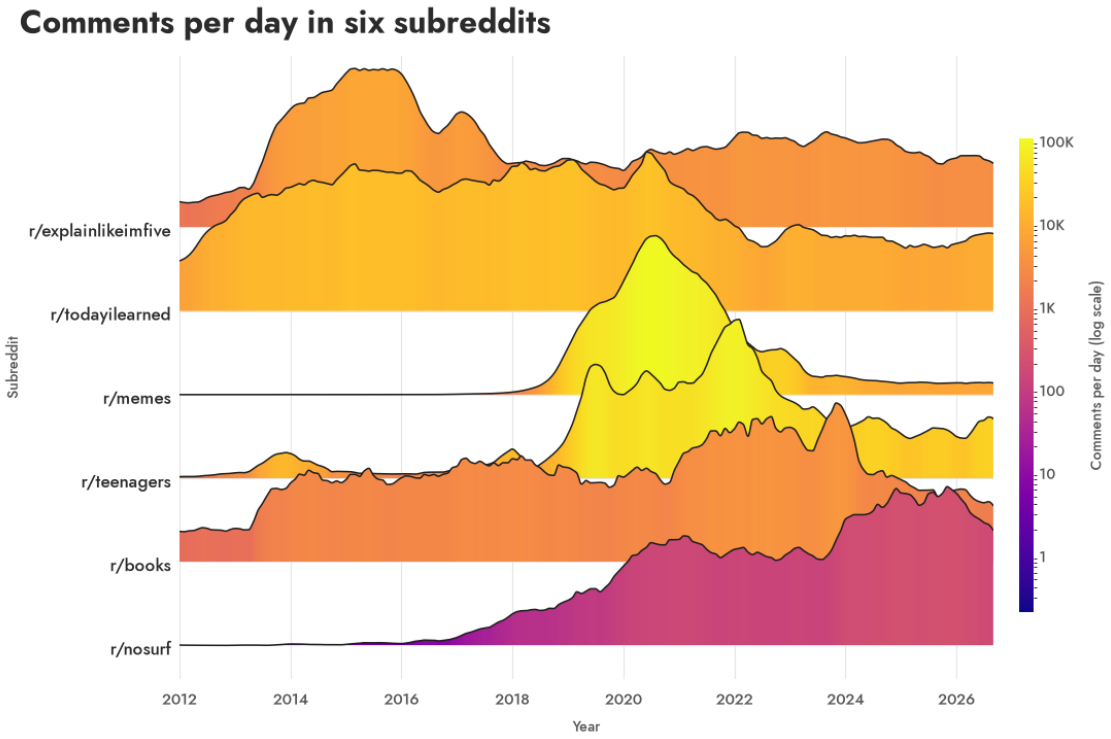


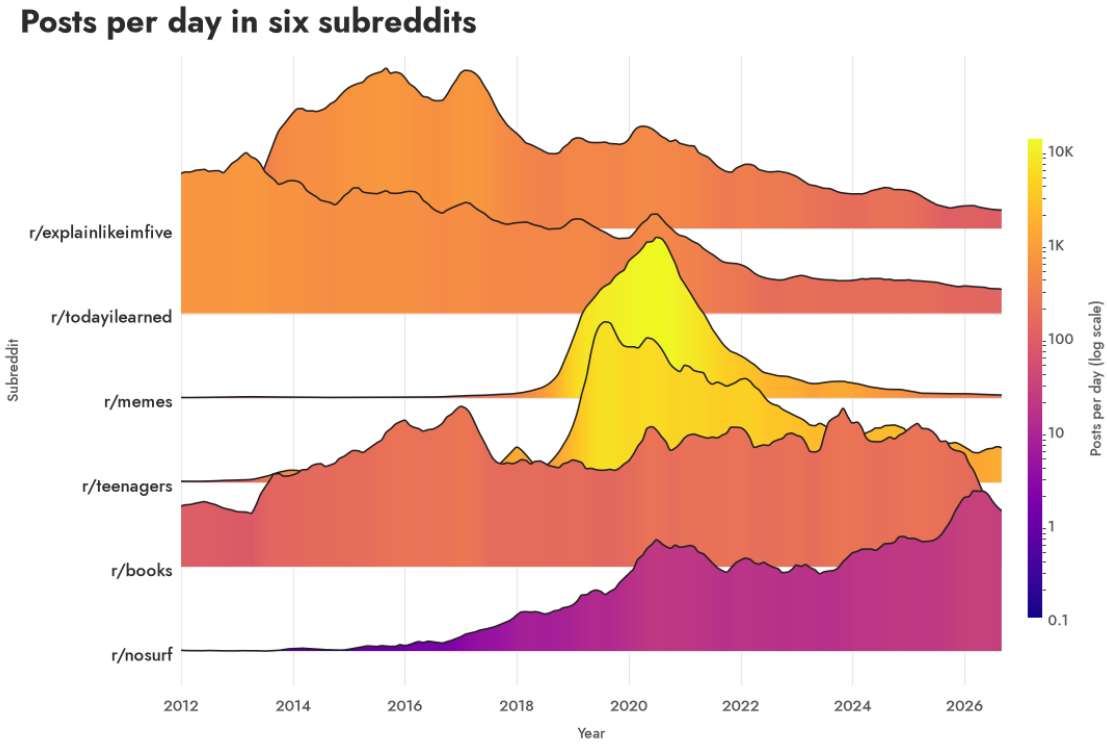
Let's take a look at stacked area chart which represents the activity of comments and posts per day for all subreddit, but in relative values.





Below is a ridgeline chart which represents the activity of each subreddit individually in gradient in absolute values.





1.3 Lexical & NLP analysis

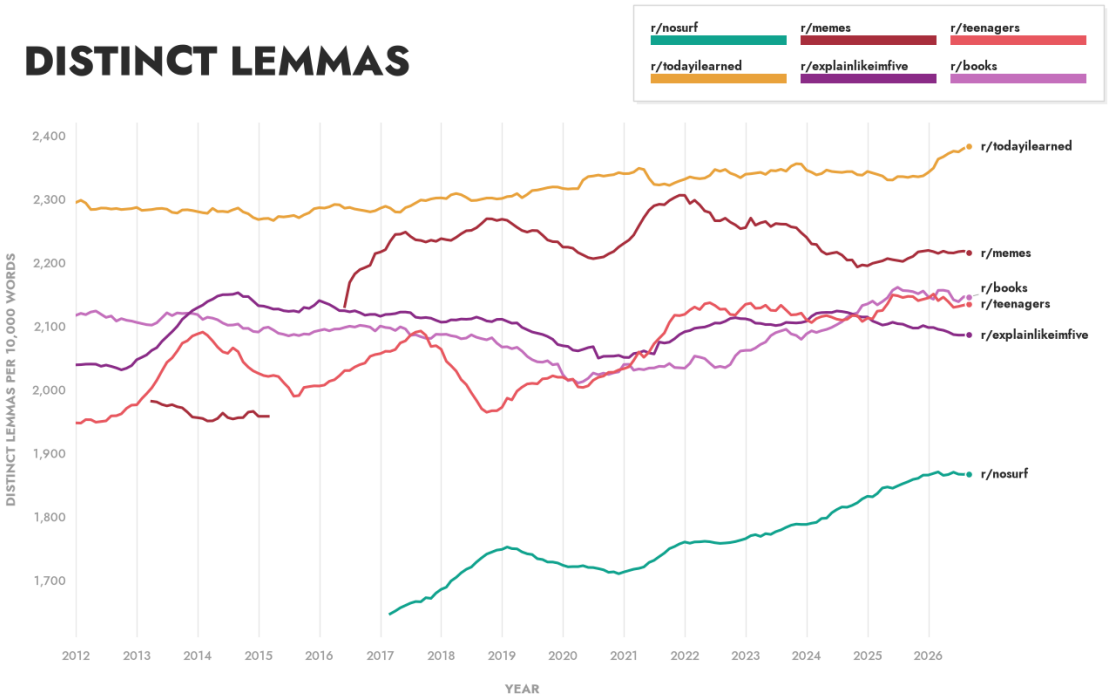
Let's analyze sentances with the following metrics:

1. Vocabulary is shrinking
2. Sentences are getting shorter
    1. Words per comment
    2. Words per sentence
3. Lempel-Ziv algorithm
    1. Concatenation of comments into fixed-size chunks per subreddit per month

1.3.1. Shrinking vocabulary

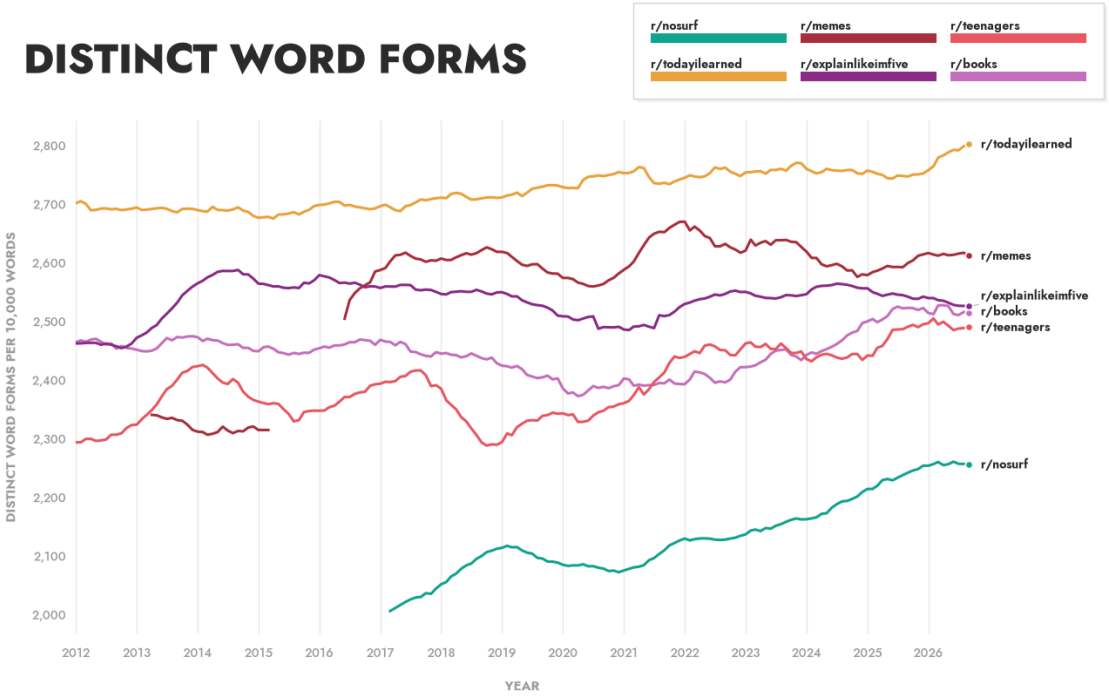
Question is if people use fewer different words over time. In order to find that we measure distinct lemmas per 10,000 words per month per subreddit.

`25,949 chunks of 10,000 words; 968 of 1062 subreddit-months have at least 3 chunks and are shown.`



And now the same measure, but for distict word forms per 10,000 word per month per subreddit.

`25,949 chunks of 10,000 words; 968 of 1062 subreddit-months have at least 3 chunks and are shown.`



1.3.2. Sentences are getting shorter

1.3.3. Lempel-Ziv algorithm

1.4. Temporal/Behavioral analysis

Let's analyze conversantions and look into: 1. Thread depth 1. Thread size 2. Share or replies vs top-level comments 3. True depth

2 H3: Wellbeing is getting worse among doomscrollers

2.1 Description of the Analysis Concept

It usually starts harmlessly — just a few minutes of catching up on short videos. But hours later, the pattern is always the same: a subtle shift in mood, a persistent dull headache, and heavy, strained eyes. Experiencing this firsthand made me wonder: *Is this just an isolated personal reaction, or is it a quantifiable, widespread digital ailment?

To answer this, I am diving into the data to uncover the hidden behavioral patterns behind screen time and well-being. Rather than forcing disparate Kaggle datasets into a single lossy table, I am treating each as a distinct chapter in a broader story. Together, they offer a multi-faceted view of how media consumption shapes human health.

2.2 Data Landscape

There are 5 such datasets:

1. Social Media Addiction dataset
2. The Dark At The End Of The Tunnel: Doomscrolling On Social Media Newsfeeds
3. Sleep & Doomscrolling Habits Dataset
4. Social Media and Mental Health
5. Student Social Media & Mental Health

2.3 Datasets Analysis

2.4 The Conceptual Framework

To synthesize these varied dimensions, I will structure the data analysis around four core well-being clusters:

- *Sleep & Circadian Rhythms:* Mapping the impact of late-night scrolling on sleep quality and sleep delay.
- *Mental Health & Affect:* Tracking mood shifts, anxiety levels, and psychological strain.
- *Addiction & Doomscrolling Behavior:* Quantifying screen time intensity, loss of agency, and compulsion loops.
- *Physical & Cognitive Well-being:* Measuring direct physical symptoms like eye strain, headaches, and cognitive fatigue.

2.4.1 Sleep & Circadian Rhythms

Picture this: You return home exhausted after a long day with zero energy left to do anything productive. Turning to your phone for a quick scroll through lighthearted reels feels like the easiest way to decompress. But before you know it, hours pass, falling asleep becomes surprisingly difficult, your sleep is fragmented with frequent night awakenings, and you wake up feeling completely drained.

It feels like a perfect vicious cycle: exhaustion drives late-night scrolling, which destroys sleep, leading to even greater fatigue the next day. But are short-form videos truly the root cause of this poor sleep quality?

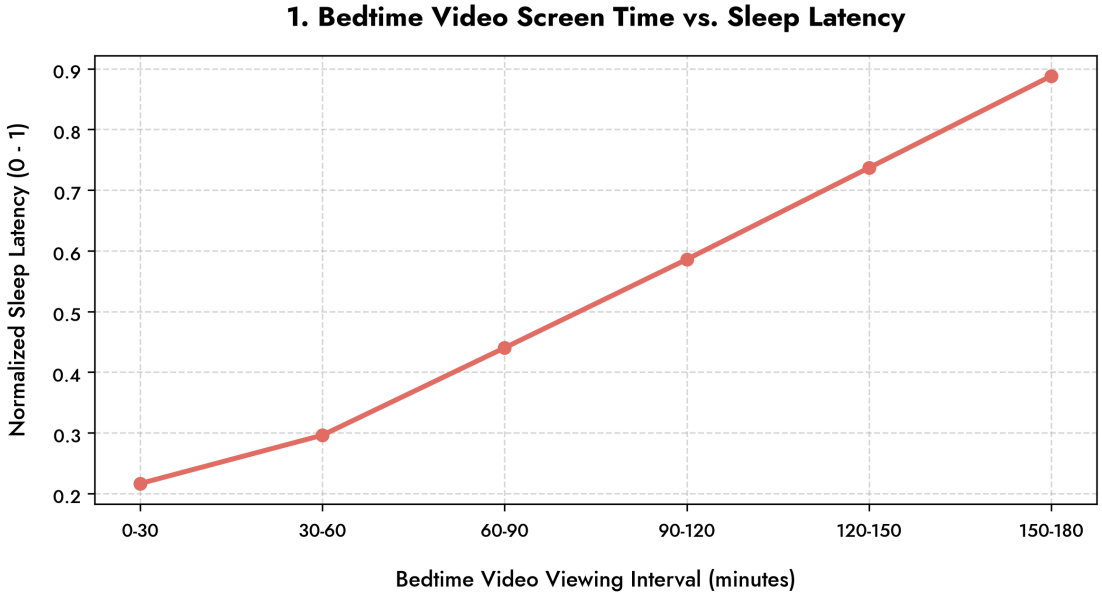
*- The relationship between the amount of time spent watching videos before bed and the time it takes to fall asleep

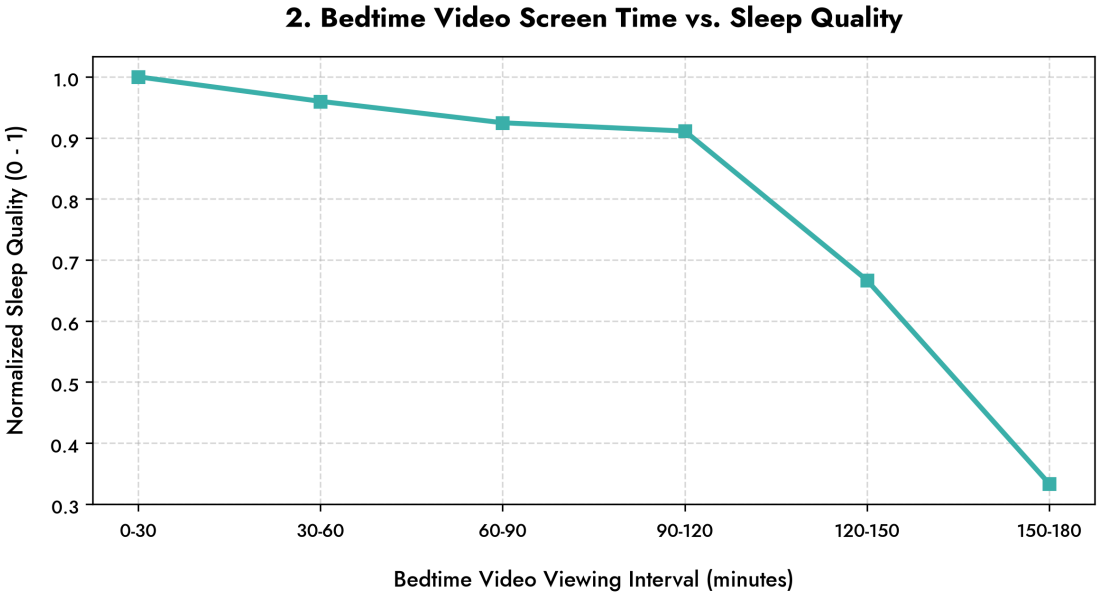
```
  screen_time_bin  avg_screen_time_min  norm_sleep_latency  \
0            0-30           18.901961            0.216460
1           30-60           44.975904            0.296132
2           60-90           72.677419            0.440153
3          90-120           99.800000            0.585965
4         120-150          122.000000            0.736842

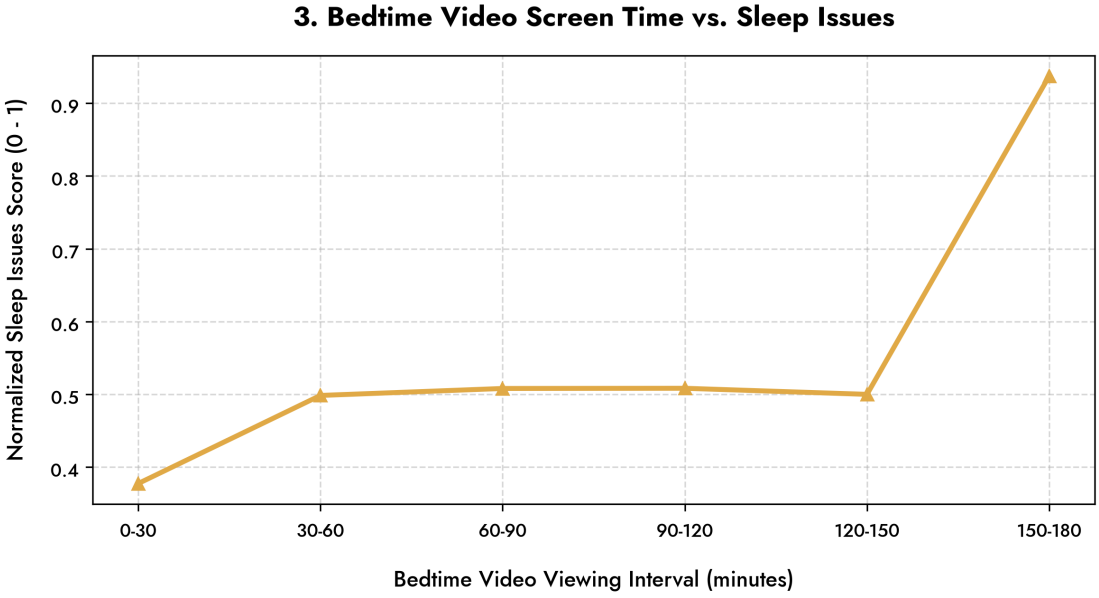
   norm_sleep_quality  norm_sleep_issues
0            1.000000           0.377451
1            0.959839           0.498494
2            0.924731           0.508065
3            0.911111           0.508333
4            0.666667           0.500000
```

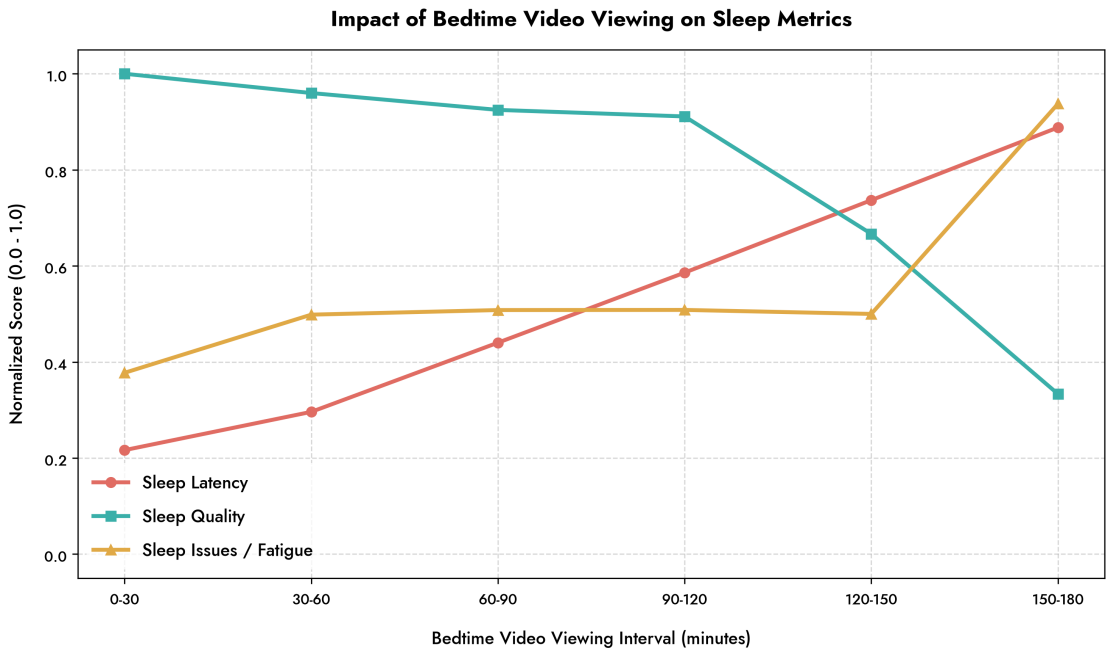
Visualisation

```
Fonts loaded successfully from: fonts
```



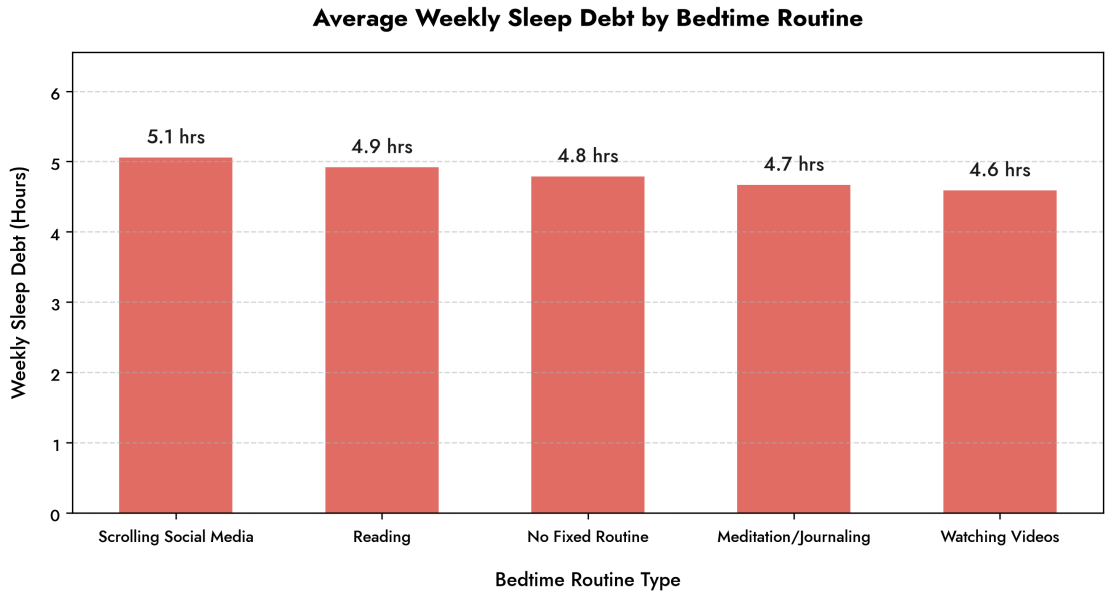

2. Bedtime Video Screen Time vs. Sleep Quality
Normalized Sleep Quality (0 - 1)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0-30
30-60
60-90
90-120
120-150
150-180
Bedtime Video Viewing Interval (minutes)


3. Bedtime Video Screen Time vs. Sleep Issues
Normalized Sleep Issues Score (0 - 1)
0.9
0.8
0.7
0.6
0.5
0.4
0-30
30-60
60-90
90-120
120-150
150-180
Bedtime Video Viewing Interval (minutes)



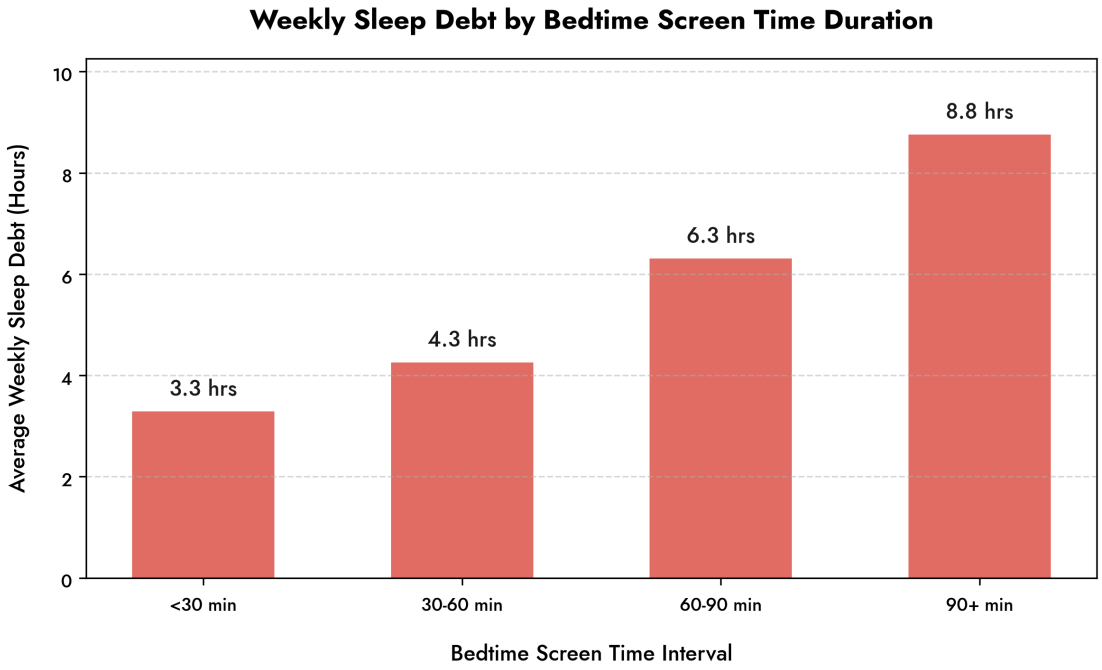
Weekly Sleep Debt Statistics by Bedtime Routine:

	bedtime_routine_type	mean	median	count
3	Scrolling Social Media	5.055696	4.20	395
2	Reading	4.917834	3.80	157
1	No Fixed Routine	4.787179	3.90	156
0	Meditation/Journaling	4.666667	3.95	108
4	Watching Videos	4.588043	3.55	184



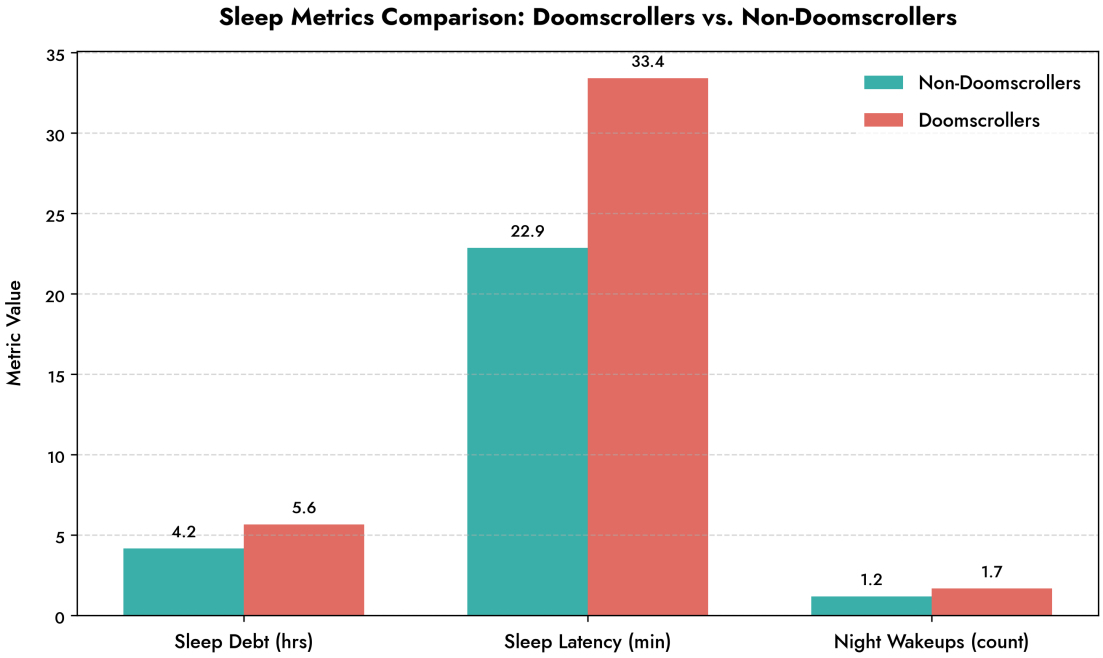
Weekly Sleep Debt by Screen Time Group:

	screen_time_group	weekly_sleep_debt_hours
0	<30 min	3.287372
1	30-60 min	4.258416
2	60-90 min	6.306736
3	90+ min	8.757273



Doomscroller Comparison Stats:

	doomscroller	sleep_debt	sleep_latency	night_wakeups	group_label
0	False	4.154771	22.854962	1.179389	Non-Doomscrollers
1	True	5.644958	33.415966	1.663866	Doomscrollers



3 H4: Crises intensify doomscrolling and negativity

3.1 Overview

This section will cover topic of H4 “crises intensify doomscrolling and negativity”.

What we currently have:

3.1.1 Events

Four major events fall inside the data range(brainrot era). We checked their dates on 2026-10-04 against news and reference sources.

1. The WHO declares COVID-19 a pandemic (11 March 2020)
2. The full-scale Russian invasion of Ukraine (24 February 2022)
3. The Hamas attack on Israel that starts the Gaza war (7 October 2023)
4. The assassination of Charlie Kirk (10 September 2025)

The last one was not on our list at first. It showed up as the busiest r/nosurf day: 11 September 2025 had 530 comments, 2.8 times the month before. Three threads carried 72% of that day, and the two biggest were about the assassination.

3.1.2 Communities

1. Short-form: r/memes + r/teenagers
2. Long-form: r/books + r/explainlikeimfive
3. Baseline: r/todayilearned

4. Qualitative: r/nosurf

We only test whether non-news communities react. Later adding news subreddits is considered.

3.1.3 Data

Comments from the six subreddits, January 2012 to September 2026. We use cleaned comments only: bots, moderator messages, templated text, removed text and spam are dropped.

3.2 Methods

List of methods we used for data exctraction:

1. Using mean VADER compound score (tone)
2. Searching words about the event itself, to check that the event reached the community
3. Searching for crisis words in general
4. `doomscroll*` mentions
5. Brainrot words (rizz, skibidi, gyatt, brain rot, tung tung, mewing)
6. Length of posting sessions

3.3 Do crises show up in what people write?

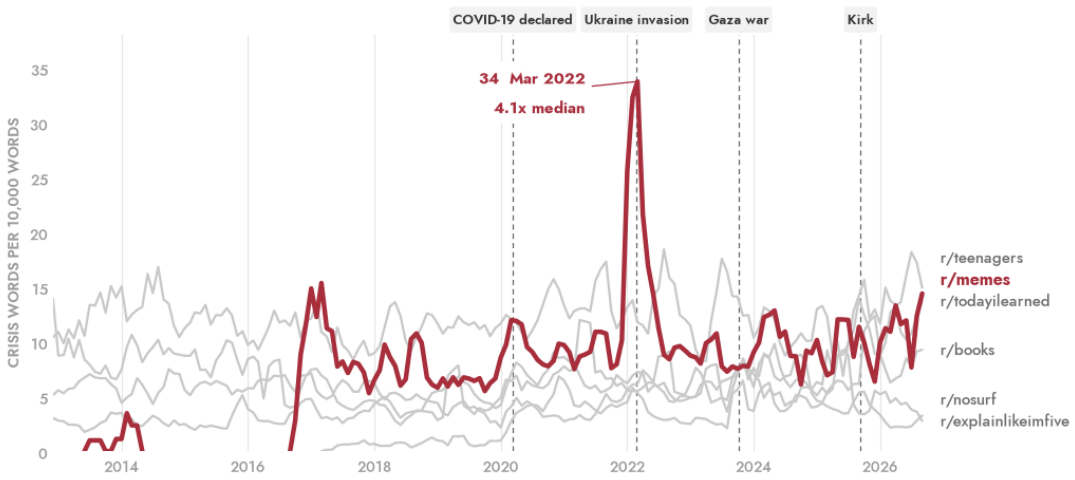
Crisis talk spikes in r/memes

What we did: Counted a broad list of crisis words per 10,000 words. The list: war, pandemic, covid, lockdown, invasion, Ukraine, Gaza, Hamas, Israel, Palestine, genocide, terror, shooting, assassination, crisis, recession, famine, refugees, earthquake, hurricane. The chart shows one line per subreddit on one scale, as a 3-month mean.

Result: r/memes shows one clear crisis peak: 34 crisis words per 10,000 words in March 2022, 4.1 times its monthly median (8.2). The other communities have no common peak:

- r/teenagers peaks in July 2026 (18.2, median 4.2)
- r/books in December 2025 (12.3, median 5.9)
- r/todayilearned in August 2022 (18.4, median 11.6)
- r/nosurf in December 2020 (6.8, median 3.1)
- r/explainlikeimfive in October 2012 (14.8, median 5.4)

Crisis talk spikes once in r/memes (Mar 2022: 34 per 10,000 words, 4.1x its median); the other communities peak at different times.



3.4 Did each event reach each community?

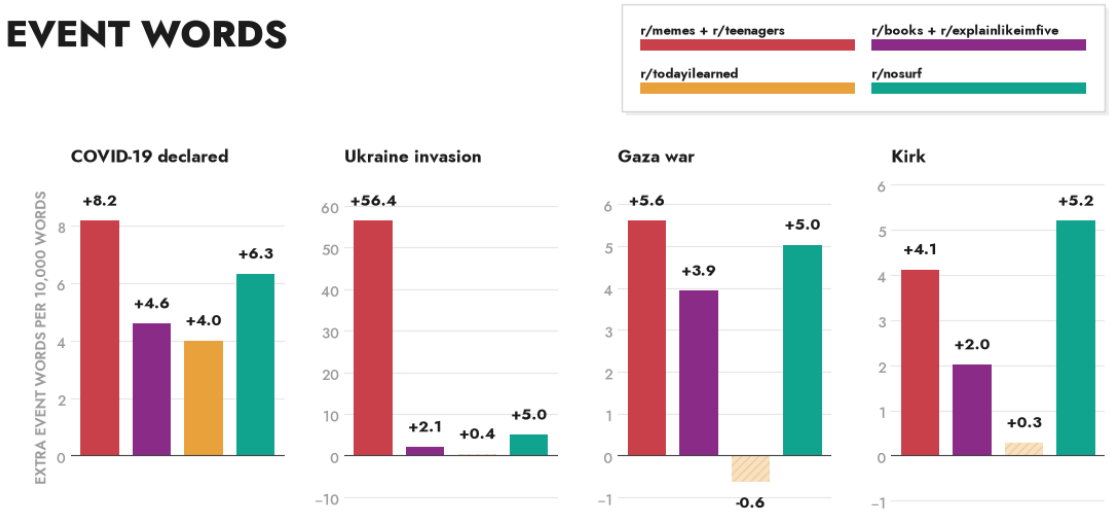
Nearly every event reached nearly every community

What we did: Counted event-specific words per 10,000 words in the 4 weeks after each event and compared them with the 8 weeks before. The words are covid/pandemic/lockdown, Ukraine/Russia/Putin/invasion, Gaza/Hamas/Israel/Palestine and Kirk/Turning Point/assassination. To judge a change we repeated the same calculation at 400 random fake dates. The share of fake dates with an equal or larger change is p. The chart shows one panel per event, and solid bars are beyond random dates ($p < 0.05$).

Result: Event words rose in 13 of 16 community-event pairs ($p < 0.05$ against random dates). The change per 10,000 words:

- Short-form: +8.2 (COVID), +56.4 (Ukraine), +5.6 (Gaza), +4.1 (Kirk)
- Long-form: +4.6, +2.1, +3.9, +2.0
- r/nosurf: +6.3, +5.0, +5.0, +5.2
- r/todayilearned: +4.0 for COVID, but not for the other three (+0.4, -0.6, +0.3)

Every event reached the communities: event words rose in 13 of 16 community-event pairs ($p < 0.05$ against random dates); r/todayilearned barely reacted to 3 of the four events.



3.5 Did tone get more negative after the events?

A small dip in tone, too weak to call a finding

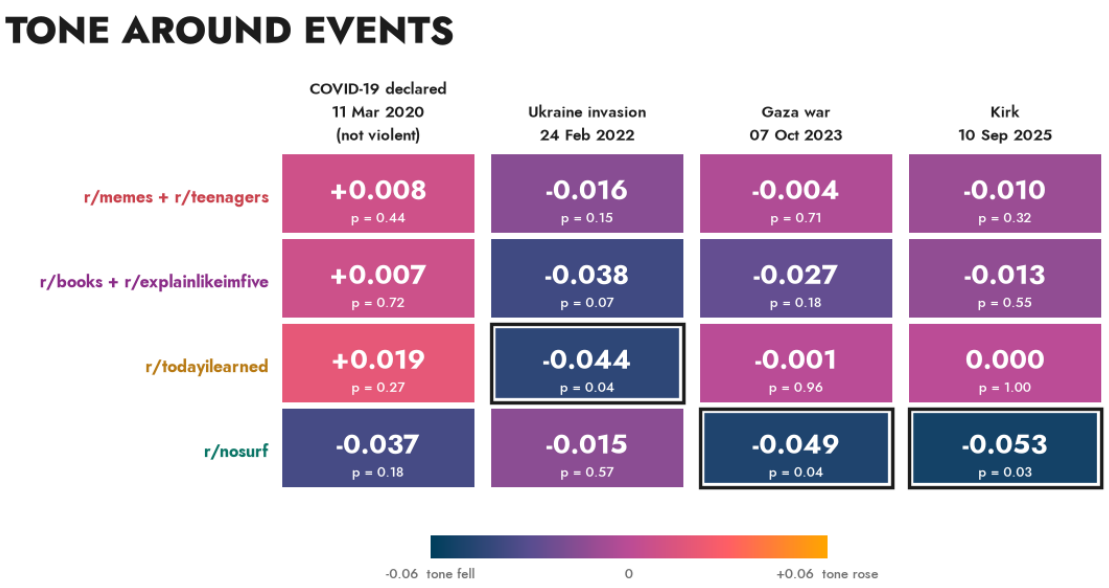
What we did: Measured the change in mean VADER compound score (-1 to +1) in the 4 weeks after each event versus the 8 weeks before. The tile matrix shows four community groups and four events: dark blue = tone fell, amber = tone rose, magenta = no change. A ring marks tiles beyond the placebo range ($p < 0.05$), and each tile shows its p-value.

Result: After the three violent events (Ukraine, Gaza, Kirk) tone was lower in 11 of the 12 community-event pairs, by up to 0.05 compound points. The twelfth, r/todayilearned after Kirk, is zero. The largest drops:

- r/nosurf: Kirk -0.053, Gaza -0.050
- r/todayilearned: Ukraine -0.044
- long-form: Ukraine -0.038

After the COVID declaration tone rose slightly in three groups (+0.007 to +0.019) and fell in r/nosurf (-0.037). Three of the 16 tiles are ringed: r/nosurf after Gaza ($p = 0.04$) and after Kirk ($p = 0.03$), and r/todayilearned after Ukraine ($p = 0.04$). About 0.8 would be expected by chance, so we do not call this a finding. For scale, r/nosurf's mean tone fell by 0.20 over nine years (0.36 in 2017, 0.16 in 2026).

After the three violent events tone dipped in 11 of 12 community-event pairs, by no more than 0.05 compound points; 3 of 16 tests (ringed) fall outside random dates, 0.8 expected by chance alone.



3.6 Did talk about doomscrolling rise after the events?

Doomscroll talk climbs, with or without crises

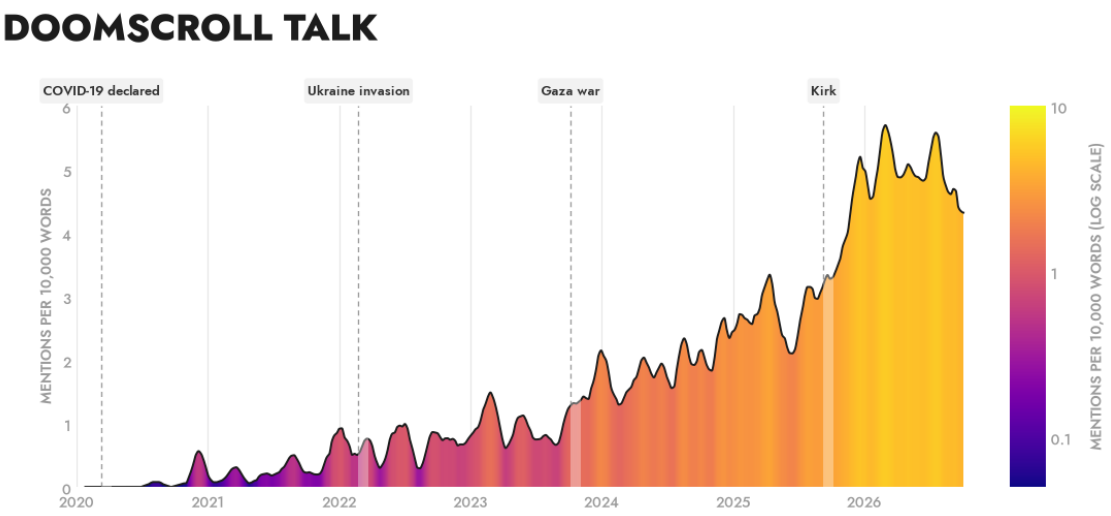
What we did: Counted `doomscroll*` and `doom surf*` mentions per 10,000 words in r/nosurf only, where we hold every comment. The rate is weekly, as a 4-week rolling rate from 2020. The chart is smoothed once more with a 5-week moving average, but the numbers below come from the unsmoothed rate. The events are dashed lines, and the 4 test weeks after each are pale bands.

Result: The rate climbs steadily. Per 10,000 words it is:

- 0.08 in 2020, 0.29 in 2021, 0.66 in 2022, 1.14 in 2023
- 1.88 in 2024, 3.24 in 2025, 4.95 in 2026 (to September)

The 4-week rate peaked at 6.3 in the week of 19 July 2026 and is 3.8 now. The four-week changes at the events are small blips on that rise: COVID 0.0 ($p = 1.0$), Ukraine +0.8 ($p = 0.05$), Gaza +0.5 ($p = 0.16$), Kirk -0.3 ($p = 0.28$), against a random-date spread of about 0.3. So the events do not explain the climb.

Doomscroll talk in r/nosurf climbed from 0.08 per 10,000 words in 2020 to 5.0 in 2026; the four events are small blips on that rise.



3.7 Did brainrot words spike around the crises?

Are brainrot spikes really tied to major events?

What we did: Counted brainrot words per 10,000 words in r/memes + r/teenagers (monthly, 3-month mean) and drew one band per word, thickness = rate. The axis starts in January 2022, because COVID-19 (March 2020) came before the slang existed. Then we pooled all the words and compared the 8 weeks before each event with the weeks after it, against random dates (every day from November 2022 that is more than 120 days from an event).

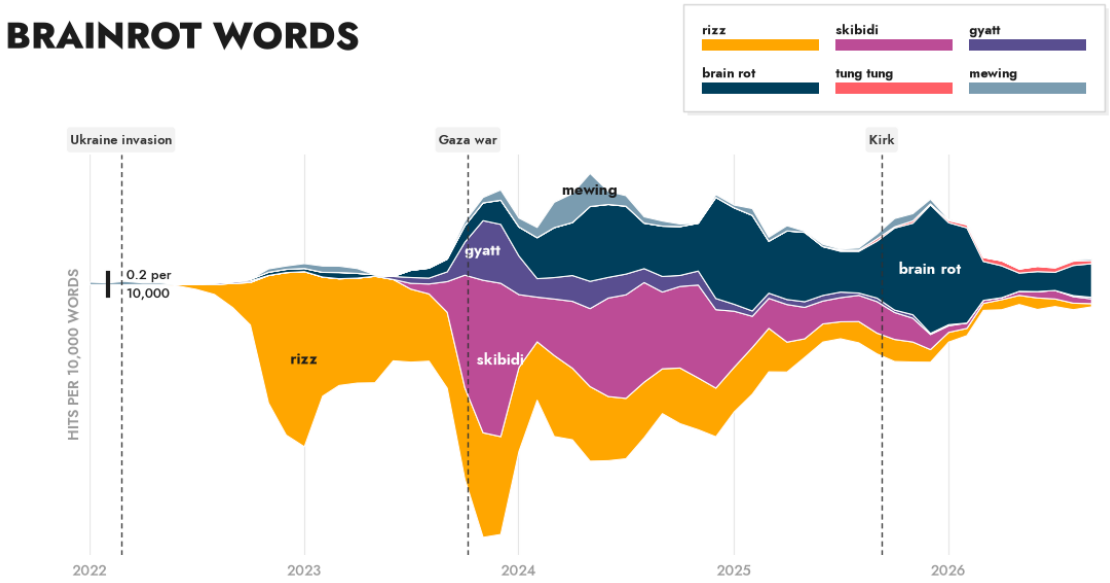
Result: In all six subreddits together, rizz is 83% of brainrot-word hits in 2022. Skibidi (23%) and gyatt (9%) join in 2023, and 'brain rot' / 'brainrot' take 41% in 2024 and 72% in 2025. The pooled rate in r/memes + r/teenagers, per 10,000 words:

- Gaza war, 4 weeks after: 0.95 before, 1.53 after (+0.58, p = 0.41), no jump yet
- Gaza war, 8 weeks after: 0.95 before, 3.40 after (+2.44), beyond every random date (they give -1.41 to +1.41)
- Kirk, 8 weeks after: 0.96 before, 1.11 after (+0.16, p = 0.69)

The jump comes late: the weekly rate is 4.45, 11.00 and 5.36 in weeks 5, 6 and 7 after 7 October 2023, against 0.43 to 1.65 in the 8 weeks before. We chose the 8-week window after seeing this weekly series, so read the p-value as a guide. With all six subreddits pooled the Gaza change is smaller (+0.38, p = 0.035). Week 6 rests on 25 sampled comments with brainrot words, from 24 authors on 3 days. The comments are not about the war: of 91 comments with brainrot words in November 2023, none mentions Gaza, Israel, Palestine or Hamas.

So brainrot words do jump after the Gaza war, five to seven weeks late and not about the war. We cannot tell whether the crisis or the slang's own rise drove it. After Kirk's killing pretty much nothing moves.

Brainrot words in r/memes and r/teenagers jump after the Gaza war and peak 6 weeks later (+2.4 per 10,000 words over the 8 weeks after, $p < 0.01$), but not after Kirk's killing (+0.2, $p = 0.69$).



3.8 Did posting sessions in r/nosurf get longer?

Posting sessions barely move, with one lead after Ukraine

What we did: Built gap-based sessions (messages by one author with gaps of at most 30 minutes) from comments and submissions, r/nosurf only. Kept sessions with 2 or more messages and split them into five length buckets (1 minute, 2-4, 5-9, 10-29, 30+). Bars show the share of sessions in each bucket: light outlined bars are the 8 weeks before, solid bars the 4 weeks after. As a yardstick we repeated the largest bucket shift at 100 random fake dates.

Result: The share of sessions with two or more messages, before vs after:

- COVID: 11.5% vs 11.4%
- Ukraine: 13.9% vs 11.9%
- Gaza: 12.5% vs 13.3%
- Kirk: 17.0% vs 16.0%

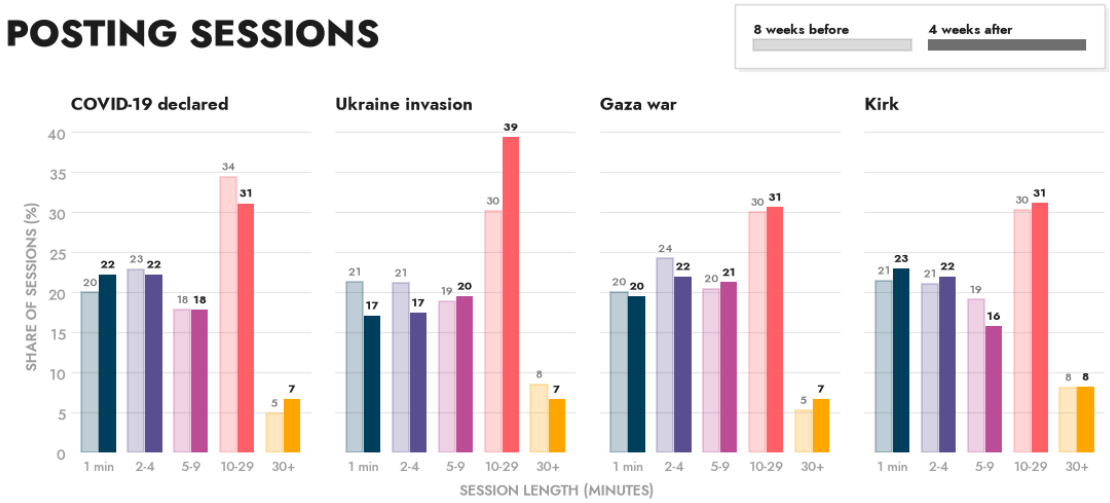
Around COVID, Gaza and Kirk no length bucket moves by more than 3.3 percentage points, and the median session stays at 6 to 7 minutes.

After the Russian invasion of Ukraine the 10-29 minute bucket grows from 30% to 39% of the sessions with two or more messages (+9.2 points), while 1-minute and 2-4-minute sessions shrink (21% to 17% each). That shift is beyond what random dates produce ($p = 0.02$). With four events tested, one p-value of 0.02 or lower happens about 8% of the time by chance, so it is a lead, not a conclusion.

So sessions are unchanged at COVID, Gaza and Kirk. Only the Ukraine invasion moves them.

Posting sessions in r/nosurf kept their usual length around three events (no length bucket moves more than 3.3 points; typical session 6-9 minutes); after the Ukraine

invasion the 10-29 min bucket grew from 30% to 39% of sessions, beyond what random dates produce (p = 0.02).



3.9 Do the same r/nosurf authors write differently after an event?

The same authors write less positively

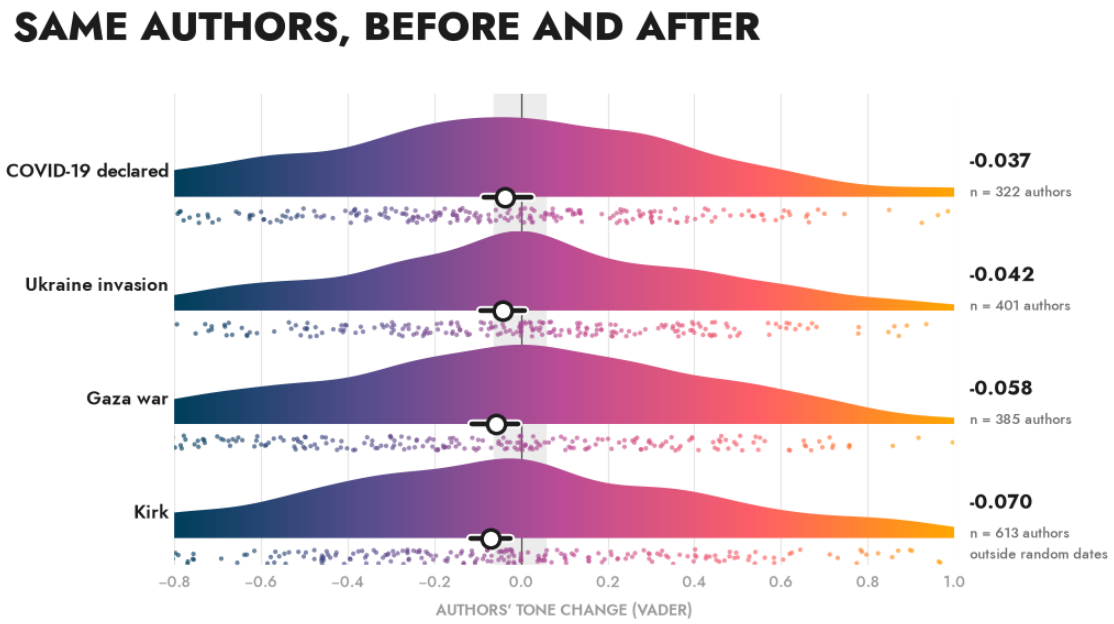
What we did: Took authors with comments in both windows (8 weeks before, 4 weeks after). Each dot is one author's change in mean VADER score, and the cloud shows the same data as a distribution. The big dot and bar are the mean with a 95% bootstrap interval, and the gray band is the 95% range of the same statistic at 120 random fake dates.

Result: The same authors wrote less positively after all four events:

- COVID: -0.037 (322 authors)
- Ukraine: -0.042 (401)
- Gaza: -0.058 (385)
- Kirk: -0.070 (613)

The intervals of Gaza and Kirk exclude zero. Only the Kirk change lies outside the random-date band (-0.063 to +0.058). This is the tone of the text, not the mood of the people.

The same r/nosurf authors wrote less positively after all four events (-0.07 to -0.04); 1 of 4 shifts falls outside random dates (Kirk).



3.10 Result

Let's pull the threads together:

- **Events reach the communities.** Event words rose in 13 of 16 community-event pairs. r/todayilearned is the exception (4.4).
- **Tone: small, consistent direction, weak evidence.** Tone was lower after the three violent events in 11 of 12 pairs, by at most 0.05. Three of 16 tests are outside the placebo range, about what we expect by chance given many correlated tests (4.5). The same r/nosurf authors wrote 0.04-0.07 less positively afterwards, with only Kirk beyond random dates (4.9).
- **Doomscroll talk climbs steadily, and the events do not explain it.** In r/nosurf the rate went from 0.08 per 10,000 words in 2020 to 4.95 in 2026. The event effects are small blips (4.6).
- **Brainrot words jump weeks after the Gaza war, not after Kirk.** The pooled rate in r/memes + r/teenagers rises from 0.95 to 3.40 per 10,000 words in the 8 weeks after the Gaza war, peaking in week 6, beyond random dates. After Kirk's killing it does not move (0.96 to 1.11). The jump comes late and is not about the war, so we cannot say the crisis caused it (4.7).
- **Posting sessions** are unchanged at three events. After the Ukraine invasion r/nosurf sessions shifted towards 10-29 minutes (30% to 39% of sessions with 2+ messages, $p = 0.02$), one of four events (4.8).

Verdict: H4 is only weakly supported. The crises do reach the communities and tone dips a little afterwards, but too weakly to call a finding. Doomscroll talk climbs steadily, with or without crises. Brainrot words jump after the Gaza war, late and not connected to event itself.

4. Result

Conclusion is not here yet, work in progress

